

Raman KHURANA



Experienced researcher with a proven track record of innovation and success. Led the publication of 4 seminal papers on search for dark matter and Higgs boson beyond the standard model, involving heavy quarks. Developed state-of-the-art transformer-based models for time series datasets. Expertise in data quality monitoring and development of tools to tackle it using low-level trigger primitives and high-level physics objects at the CMS experiment at CERN.

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ACADEMIC EMPLOYMENT

- **Staff Researcher**, Northwestern University, USA. September 2022 - Now
 - **Supervisor:** Prof. Diego Kalbjan
 - **Main projects:** Development of an Automated Feature Engineering model for streaming data.
 - Enhancing the capabilities of existing transformer and MLP based time series forecasting model to handle wide data sources; static and temporal covariates.
 - Development of real time streaming pipeline to make predictions using resource constrained edge-devices like FPGA/Raspberry Pis.
 - Development of pipeline making real time prediction data streaming from sensors or IoT devices.
- **Research Associate**, National Taiwan University. April 2022 - September 2022
 - **Supervisor:** Prof. George Wei-Shu Hou
 - **Main projects:** Development of data driven background estimate and upper limit estimation framework to search for BSM Higgs boson involving same sign top quarks events in association with charm quark.
- **Post-doctoral Researcher**, Florida State University, USA. September 2018 - March 2022
 - **Supervisor:** Prof. Andrew Askew
 - **Main projects:** Extending the search for dark matter using Higgs boson decaying to bottom quarks incorporating machine learning techniques.
 - Expanding the experimental reach of the dark matter search by combining five mono-H final states.
 - Spin off the mono-H (bb) as search for dark matter in non-resonant bottom quarks and p_T^{miss} targeting complimentary phase space.
 - Responsible for operations of CMS Electromagnetic Calorimeter (ECAL) and improving the ECAL L1 trigger primitive generation.
- **Post-doctoral Researcher**, National Central University, Taiwan, March 2015 - September 2018
 - **Supervisor:** Prof. Shin Shan Yu (Eiko)
 - **Main projects:** Search for dark matter using Higgs boson decaying to bottom quarks.
 - Search for new heavy resonances in VH(bb+MET) and HH (bbbb) final state
 - Jet and p_T^{miss} data quality monitoring (DQM) group leader.
 - Exploiting the timing capabilities of the HGCal Silicon sensor using test beam data.
- **Research Fellow**, Saha Institute of Nuclear Physics, Kolkata, India, July 2014 - February 2015
 - **Supervisor:** Prof. Sunanda Banerjee
 - **Main projects:** Standalone simulation of the HGCal detector, developing the validation software for multiple steps of during reconstructions of the signal from HGCal sensors.

EDUCATION

- PhD, Experimental High Energy Physics, University of Calcutta and Saha Institute of Nuclear Physics, India, 2014.
 - **Thesis supervisor:** Prof. Satyaki Bhattacharya.
 - **Thesis Title:** Search for Higgs boson in the $H \rightarrow 2l2\tau$ final state using the data collected at CMS experiment of LHC.
- MSc, Physics (Electronics Specialization), University of Delhi, India, 2009.
 - **Thesis supervisor:** Prof. Vinay Gupta and Prof. Amitabha Mukherjee
 - **Thesis Title:** Developed a SONAR like table top experiment to **demonstrate the triggering and data acquisition system (DAQ)** using ultrasonic sensors, 555 timer and μ -controller and display the location of target on an interfaced LCD.
- BSc, Physics (Hons), University of Delhi, India, 2007.

AWARDS AND FELLOWSHIPS

- Awarded with Ramanujan Fellowship 2021 with a research grant of \$150K, to carryout the dark matter search project + CMS detector upgrade.
- Referee for Journal of High Energy Physics (JHEP)
- Referee for Data Science Journal (DSJ)
- Senior Research Fellowship, Awarded by CSIR, 2011
- Junior Research Fellowship, Awarded by CSIR, 2009
- Qualified National Eligibility Test for lectureship, Conducted & Awarded by CSIR, New Delhi

SKILL-SET

Programming languages	C, C++, python, shell scripting
HEP Tools	uproot, awkward arrays, ROOT, RooFit, RooStats, combine
Visualisation	ROOT, matplotlib, seaborn
ML/AI tools	scipy, keras, tensorflow and sk-learn, pytorch.
Big data Processing	github CI/CD, docker, containerisation, kubernetes deployment
Softwares	GEANT4, PYTHIA, MADGRAPH
Operating system	Linux, Macintosh, Windows.

RESEARCH HIGHLIGHTS

AI/ML Model development In a recent project, I developed an innovative deep learning model using PyTorch, specifically designed for multivariate forecasting. This model adeptly handles both temporal and static datasets, setting a new benchmark. My approach involved comprehensive comparative analysis across various standard time series datasets, ensuring the model's robustness and versatility.

To achieve optimal performance, I orchestrated an extensive hyperparameter optimization process using a Kubernetes cluster, strategically configured from scratch with three nodes. This infrastructure not only supported the computational demands of the project but also fostered an efficient, scalable environment for model tuning.

A key component of this project was the integration of GitHub CI/CD pipelines. This setup was crucial for the automatic creation and management of Docker images, streamlining the deployment process and enhancing the overall workflow efficiency.

Moreover, I developed and deployed custom scripts using cron jobs. These scripts were ingeniously designed to trigger the optimization process automatically whenever updates were made to the GitHub repository. This proactive approach not only saved time but also ensured that the model continually evolved with the latest advancements.

This project stands as a testament to my skills in machine learning, software development, and systems engineering, demonstrating a deep understanding of both the theoretical and practical aspects of advanced predictive modeling.

Independent Research

I have played a crucial role as the lead analyst and driving force behind the publication of numerous search and review analyses. My primary focus has been on the search for the Higgs boson and its association with dark matter. Below, I present several key contributions that I conceived and successfully executed. I presented these analyses at multiple occasions of pre-approval/approval/international conferences. Full details are presented after publication for reference.

- Most deep learning models are unable to use leverage the power of static data and recent experiments have proved its usability and importance. I am actively heading the assessment of existing deep learning time series forecasting models to uncover weaknesses and devise solutions (work in progress). This involves reviewing the existing novel transformer variant models and extend them via various embedding techniques to use the static meta-data to enhance the model prediction performance github. I also supervised two team of MS (in Analytic) students who developed techniques for anomaly and drift detection in time series data github.
- I am leading and supervising a team of Master students to develop a real time prediction pipeline using edge devices (Raspberry-pi) to reduce time delay to send data to a central cluster for processing and prediction. The setup consist of five devices forming a network and streaming wild-fire related data from IoT sensors and making prediction about probability of wild fire next day. github
- In 2014, I initiated and led groundbreaking dark matter search using the Higgs + p_T^{miss} events, resulting in two published works featured in **Eur. Phys. J. C 79 (2019) 280** and **JHEP 10 (2017) 180**. The initial publication harnessed a small portion of the 13 TeV dataset (2.3 fb^{-1}) to explore potential theoretical models, conduct feasibility assessments, and perform simulation studies for the mono-H (bb) final state, characterized by the highest branching ratio. I developed the analysis framework from the ground up, employing C++ classes and adhering to object-oriented programming principles. The first findings were subsequently merged with the $\gamma\gamma$ final state analysis to comprehensively cover the entire phase space. With an increase in luminosity, we probed novel models proposing the mono-H signature and applied machine learning techniques to enhance the discovery potential. The refined outcomes culminated in a publication in **Eur. Phys. J. C 79 (2019) 280**, establishing the most stringent limits on the production of dark matter in association with the Higgs boson.
- As data collection from CMS expanded, enabling broader exploration of the mono-Higgs events across an extended phase space, I took charge of an initiative to synchronize teams engaged in the study of other Higgs boson decay modes. I led this project, performing the statistical combination of all five decay modes, culminating in the publication of **JHEP 03 (2020) 025**.
- For my PhD thesis I worked on a unique decay to search for Higgs boson at the CMS. I searched for heavy Higgs boson with $m_H > 200 \text{ GeV}$ in the $ZZ (\rightarrow 2l2\tau)$ final state. It become as documented in **JHEP 03 (2012) 081**, **JHEP 01 (2013) 063**.

Convener of Jet and p_T^{miss} data quality monitoring (2016 to 2018) I led a team of 12-15 physicists to timely perform quality checks of the data collected during a very crucial period, when CMS collected the biggest chunk of data ($\approx 100 \text{ fb}^{-1}$). The duties and responsibilities were mainly:

- To check the quality of collected data using Jets and missing transverse momentum (p_T^{miss}) related variables.
- Validate the changes in the software using simulated and real data events.
- Timely detection of the problems and bring them to the attention of colleagues.

- Improve the data certification process to ensure good quality of data without loss of time and data itself.
- Developed and deployed a new tool for Jet and p_T^{miss} DQM to see the time evolution of the jet energy fraction and p_T^{miss} variables.

Member of analysis review committee The analysis review committee is usually a team of 4-5 members from the collaboration with expertise in one or more areas. The members review all steps of an analysis and scrutinize the results presented in the potential paper draft. I have reviewed four analyses targeting physics beyond the standard model (SM). **CMS-EXO-20-005, PRD 97, 072006 (2018), JHEP 07 (2018) 075, CMS-B2G-16-021.**

Reviewer of b-tagging usage An analysis destined for publication has to pass the object review. I reviewed b-tagging usage for > 30 analyses in the CMS Exotica group during 2015 to 2017. This is very critical step to make sure that b-tagging algorithm, working point usage, systematic uncertainties, scale factors to account for data to simulation efficiency difference are used properly and documented in the paper or public document.

Hardware contributions My major contribution to hardware are in calorimeters and are listed below:

- Performed the component testing of the HCAL backend electronics for the forwards HCAL upgrade. The $\mu - HTR$ tested already collecting data since 2015.
- Created the very first standalone simulation setup for the High Granular Calorimeter (HGCAL) and performed the early simulation study of the active materials, resolution and Moliere radius.
- The test beam data of one module of HGCAL was used to establish that the time resolution of the Silicon (Si) sensors can be used to separate hard interaction from pileup interactions **JINST 13 (2018) 10, P10023.**
- I was responsible to measure the relative contribution of the effects leading to the ECAL L1 trigger pre-firing, observed in Run 2. The project is being extended to possibly emulate the effect in simulation and also control and monitor in the Run 3. **(Paper in preparation).**

Other contributions My other contribution to CMS Experiment includes (but not limited to):

- Hadronic τ Physics Object Group: With the increase in instantaneous luminosity resulting in high pileup came with degraded hadronic tau identification efficiency. I tuned and optimised the selection for tau ID algorithm to mitigate the pileup effects and make it almost independent of pileup. Later, I identified the source of low reconstruction efficiency for three prong taus and provided partial solutions. I concluded my contributions to tau POG with electron to τ and μ to τ mis-identification rate using template based tag and probe technique. The findings were published in **JINST 11 (2016) P01019**

MENTORING EXPERIENCE

- **CMS DAS 2021 at Fermilab, USA:** Facilitator/tutor for hadronic tau τ_h reconstruction and $Z \rightarrow \tau\tau$ cross-section measurement in 2 week virtual CMS Data Analysis School (CMS DAS) organised by Fermilab.
- **Mr. Deepak Kumar, PhD, IISc, Bangalore.** Thesis project: Search for dark matter produced in association with Higgs boson decaying into bottom quarks using full Run 2 data. Status: Completed.
- **Mr. Praveen Chandra Tiwari, PhD, IISc, Bangalore.** Thesis project: Search for dark matter produced in association with non-resonant production of pair of bottom quarks. Status: Completed.
- **Ms. Pampa Ghose, PhD, FSU, USA.** Project: Study of ECAL L1 Trigger pre-firing using simulation data. Status: Completed.
- **Mr. Kung-Hsiang, master student, National Central University.** Thesis: Study of substructure variable for mono-Higgs $\rightarrow bb$ analysis. Status: Completed.
- **Ms. Fasya Khuzaimah, master student, National Central University.** Thesis: Study of double b-tagger variable and derivation of data to simulation scale factors for mono-Higgs $\rightarrow bb$ analysis.
- **Mr. Ching-Wei Chang, master student, National Central University.** Thesis: Search for new resonance decaying into pair of Higgs boson ($HH \rightarrow 4b$). Status: completed.
- **Ms. Fang-Ying Tsai, master student, National Central University.** Thesis: Search for dark matter produced in association with Higgs boson decaying into bottom quarks (mono-Higgs $\rightarrow bb$). Status: completed.

INVITED PRESENTATIONS AT INTERNATIONAL CONFERENCES

Invited/Plenary/Parallel Presentation

- Search for Dark Matter using Higgs boson as a tool. ICHEP 2022.
- Search for dark matter at CMS, Virtual World, ICHEP 2020.
- Searches for dark matter and long-lived particles at the LHC, Cancelled due to Covid 19, La Thuile, Italy, 2020.
- Dark Matter searches at LHC, LHCP2017, Shanghai, China.
- Dark Matter searches at LHC, Warsaw Workshop on Non-Standard Dark Matter 2016, Poland.
- Search for heavy resonances in the W/Z-tagged dijet mass spectrum at CMS, PSROC 2016, Kaohsiung city, Taiwan.
- Search for standard model Higgs boson in tau final states, Workshop on Higgs and BSM physics 2013, ICTP, Italy.

Posters

- Investigation of the fast timing capabilities of the Silicon sensors for the CMS high granularity calorimeter, LHCP 2017, Shanghai, China.
- Identification of hadronic tau decays in CMS, PANIC 2014, Hamburg, Germany. **JINST 11 (2016) P01019.**
- Search for Standard Model Higgs Boson in $H \rightarrow ZZ \rightarrow 2l2\tau$ decay channel with CMS detector at $\sqrt{s} = 7$ and 8 TeV, EPS HEP 2013, Stockholm.
- Tau reconstruction and identification at CMS, ICHEP 2012, Australia. **PoS ICHEP2012 (2013) 518**

- Tau identification commissioning results, Lepton Photon 2011, Mumbai, India. **Pramana 79 (2012) 1235-1238**

Workshop Participation

- ECAL L1 Trigger pre-firing studies. ECAL days in Zurich 2019, Zurich, Switzerland.
- Search for dark matter produced in association with Higgs boson using Run 2 data., CMS Exotica Workshop, Greece.
- Search for dark matter in Higgs ($H \rightarrow b\bar{b}$) + p_T^{miss} final state, CMS Exotica workshop 2015, Venice, Italy.

PUBLICATIONS

I am a member of the CMS Collaboration and by convention because of my continuous service towards the collaboration an author of more than 1000 publications . I list here only those in which either I was leading the analysis/study or I have contributed significantly to the publication.

1. Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s}=13$ TeV, **CMS-PAS-SUS-23-008**
2. Search for new Higgs bosons through same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s} = 13$ TeV. **Phys. Lett. B 850 (2024) 138478**
3. Search for dark matter particles produced in association with a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV. **JHEP 03 (2020) 025**.
4. Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s} = 13$ TeV using the CMS detector **JINST 14 (2019) P07004**.
5. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton-proton collisions at $\sqrt{s}=13$ TeV. **Eur. Phys. J. C 79 (2019) 280**.
6. Search for a massive resonance decaying to a pair of Higgs bosons in the four b quark final state in proton-proton collisions at $\sqrt{s} = 13$ TeV,. **Phys. Lett. B 781 (2018) 244**.
7. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV. **JHEP 11 (2018) 172**.
8. First beam tests of prototype silicon modules for the CMS High Granularity Endcap Calorimeter **JINST 13 (2018) 10, P10023**.
9. Search for associated production of dark matter with a Higgs boson decaying to $b\bar{b}$ or $\gamma\gamma$ at $\sqrt{s} = 13$ TeV. **JHEP 10 (2017) 180**.
10. Search for heavy resonances decaying into a vector boson and a Higgs boson in the $(\nu\nu, \nu, ll) b\bar{b}$ final states. **Phys. Lett. B 768 (2017) 137**.
11. Reconstruction and identification of τ lepton decays to hadrons and ν_τ at CMS. **JINST 11 (2016) P01019** .
12. Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons. **JHEP 10 (2015) 144**.
13. Properties of the Higgs-like boson in the decay $H \rightarrow ZZ \rightarrow 4l$ in pp collisions at $\sqrt{s} = 7$ TeV and 8 TeV. **Phys. Rev. D 89 (2014) 092007**.
14. Evidence for the direct decay of the 125 GeV Higgs boson to fermions **Nature Physics 10 (2014) 557**.
15. Measurement of the W^+W^- and ZZ production cross sections in pp collisions at $\sqrt{s} = 8$ TeV. **Phys. Lett. B 721 (2013) 190**.
16. Measurement of the ZZ production cross section and search for anomalous couplings at $\sqrt{s} = 7$ TeV. **JHEP 01 (2013) 063**.
17. Search for a standard-model-like Higgs boson with a mass in the range 145 to 1000 GeV at the LHC. **Eur. Phys. J. C 73 (2013) 2469**.
18. Updated results on the new boson discovered in the search for the standard model Higgs boson in the $H \rightarrow ZZ \rightarrow 4l$ channel in pp collisions at $\sqrt{s}=7$ and 8 TeV. **Phys. Rev. Lett. 110 (2013) 081803**.
19. Search for a Standard Model Higgs boson produced in the decay channel $H \rightarrow ZZ \rightarrow 2l2\tau$ with CMS detector at $\sqrt{s}=7$ TeV,. **JHEP 03 (2012) 081**.
20. Dark Matter Benchmark Models for Early LHC Run-2 Searches: Report of the ATLAS/CMS Dark Matter Forum. **Phys. Dark Univ. 26 (2019) 100371**.

DETAILED RESEARCH EXPERIENCE

I have worked on many different but related projects throughout my research career. In the CMS collaboration, the group/team which provides the objects such as, electrons, muons, jets, hadronic taus etc is called a Physics Object Group (POG) and the group coordinating the analyses is called a Physics Analysis Group (PAG). The POG and PAG are lead by usually two physicists who are expert in the subject, called conveners. I have worked with many different physics objects and physics analysis groups. Here is the summary of my research experience:

Physics Analysis

Search for Dark Matter candidate I proposed an interesting search for the dark matter produced in association with Higgs boson (mono-Higgs) and lead all the publications including mono-H $\rightarrow$ bb final state. I am **principle analyst and coordinator** for dark matter in mono-Higgs ($\rightarrow$ bb) final state. **[2015-Now]**. I have developed the analysis framework starting from MINIAOD including analysis strategy, statistical analysis, and interpretation of results for the mono-Higgs ($\rightarrow$ bb) final state when the Higgs boson is Lorentz boosted and decays in into a pair of b-jets close to each other. I have continuously worked towards bringing new ideas and improving the analysis using new techniques which has resulted in the present shape of the analysis.

- I have performed the first statistical combination of mono-Higgs analyses result in $\rightarrow$ bb and $\rightarrow \gamma\gamma$ decay channels using 2.3 fb^{-1} of pp collision data. **JHEP 10 (2017) 180**.
- I have developed novel techniques to reconstruct and tag the Lorentz boosted Higgs boson ($\rightarrow$ bb) and performed the analysis with $\approx 15\times$ more data collected at CMS. **Eur. Phys. J. C 79 (2019) 280, JHEP 11 (2018) 172**.
- I later performed the first statistical combination of mono-Higgs analyses in five decay channels, namely, bb, $\gamma\gamma$, $\tau\tau$, ZZ, WW. **JHEP 03 (2020) 025**. This was very complex analysis due to analysis performed by different teams and many correlated nuisances to deal with.
- Given the huge amount of data to analyze and new analysis softwares and techniques are available. I have recently developed the analysis framework using NANOAO framework which will be new norm in near future. The analysis is now performed in fully parallel and columnar manner using new tools like awkward1 arrays, uproot4, and more general python tools like pandas and matplotlib for cleaning and visualisation.
- The fore mentioned developments are now being used to perform the full Run 2 version of the analysis of mono-Higgs. In order to add interpretation which predicts low and medium Lorentz boost for the Higgs boson a new category is being worked out, where Higgs boson can be reconstructed two jets. I have already demonstrated that resolved category captures the lost signal w.r.t analysis with merged boosted Higgs boson alone. Performed the statistical combination of low and high Lorentz boosted Higgs boson categories to gain sensitivity for the full CMS Run-2 analysis. (In progress).
- The resolved category added to the mono-Higgs can be tweaked to search the dark matter produced in association with non-resonant pair of bottom quarks. I have developed the analysis strategy and tools for the same. The analysis is advance stage and planed for Moriond 2021 or soon after. (In progress).

Search for Heavy Resonances Another related area of interest is to search for new heavy resonances which are signature of physics beyond the SM. I consider the final states when a heavy resonance decays into a pair of Higgs bosons (HH) or a Higgs boson in association with a weak vector boson (VH).

- Developed the strategy for the signal event (HH $\rightarrow$ 4b) selection and optimised it for better signal significance. **Phys. Lett. B 781 (2018) 244**
- Member of the team optimising the signal (VH $\rightarrow ll, \nu\nu + bb$) selection and performing the cross-check as a parallel analysis. **Phys. Lett. B 768 (2017) 137**

Search for Higgs boson The research work accomplished during my PhD was the search for the Higgs boson in ZZ $\rightarrow 2l2\tau$ final states. The analysis itself consisted of 8 decay modes. The huge background and many decay modes made it one of the most complex analyses.

- Performed the feasibility study and developed the trigger strategy.
- Developed independent tools for event selection, optimisation and statistical analysis.
- The improvements made in the hadronic tau reconstruction directly used in the analysis to improve the significance.
- The analysis was published multiple time independently and as a part of H $\rightarrow$ ZZ $\rightarrow$ 4l analysis. **JHEP 03 (2012) 081, EPJC 73 (2013) 2469,**
- Results were later also used to measure the production cross-section of SM diboson process. **JHEP 01 (2013) 063.**

Physics Objects

Jet and p_T^{miss} The group's aim is to understand the fundamental physics objects jets and p_T^{miss} . These are key part in most from the searches hunting for dark matter, graviton, SUSY searches etc. I have been studying the high p_T^{miss} tails, which is generally the signature for signal, via multiple methods for a significant fraction of my research career.

- **Detector defects:** The presence of dead electronics channels in the Electromagnetic calorimeter (ECAL) and Hadron calorimeter (HCAL) can lead to high fake p_T^{miss} . The study carried out to produce the map of these detector holes and an algorithm was developed with the detector performance group to recover the energy deposited in this region. The new algorithm was implemented and validated using data collected in 2015.
- **p_T^{miss} Scanning:** Another source of high fake p_T^{miss} in data is due to detector electronics defects or unknown features in the reconstruction algorithms. These effects are usually overlaid in the newly collected data and be can easily be seen in p_T^{miss} tails. In the study performed two measures were taken. First, the events with high and exceptionally high values of p_T^{miss} were checked manually and to trace the actual problem. Second, a new data stream (Hotline) was used to check the tails and look for possible issues. Hotline data stream saves events with very high energy objects and ready for analysis within couple of hours of data taking. Finally, the filters were developed and performance was checked regularly in data. These filters are now recommended for collaboration wide usage and useful in rejecting huge filters are further tuned after better understanding of data collected p_T^{miss} tails.
- **Jet and p_T^{miss} Data Quality Monitoring:** Software expert for the JET- p_T^{miss} DQM and validation from 2016-2018. The software was updated and upgraded from time to time and more functionalities were added to spot problems related to detector, reconstruction and for debugging.
- **AUXJMEDQM:** I developed a new tool (AUXJMEDQM) to see the time evolution of observables. The newly developed tool uses the existing information which are produced to monitor the data quality and check the time evolution of a given variable for the full dataset in a short duration. The tool is mainly a quick but robust outlier detector. The changes in the detector configuration is spotted effortlessly. It is welcomed and now part of the monitoring producere to see the time evolution distributions from time to time.

The studies performed in the Jet p_T^{miss} group were part of the **JINST 14 (2019) P07004**

Tau

Tau POG is charged with the understanding hadronic decays of taus, reconstruct and isolate them. They are key part of many searches, including $H \rightarrow \tau\tau$, SUSY and SM measurements. My contribution to Tau POG has been multi-fold.

- Measured the tau energy scale and its uncertainty using the tau-like gluon jets.
- I setup the first version of Tau ID Validation software to keep track of performance with changes and time.
- Devised the technique to make tau decay mode reconstruction to be pile up independent.
- Optimised the selection for three prong tau decay modes which lead to increase in signal efficiency by 10% keeping the mis-identification rate at same level.
- The reconstruction of high p_T hadronic tau was improved by optimising the selection which resulted in 40-50% increase in the efficiency above 200 GeV while keeping low mis-identification rate.
- The rate of mis-identification from electron to hadronic tau was measured to be 2-5% depending on the selection criteria used. The measurement was done using Tag and Probe method.

The studies performed in the Tau group were part of the journal published paper **JINST 11 (2016) P01019** and also recommendation for collaboration wide usage.

Detector

ECAL

During my post-doctoral research period at FSU I am a member of the ECAL operations team and perform a variety of studies for ECAL Trigger group. I am an expert and experienced CMS ECAL shifter with role of DGL/Detector On Call [DGL/DoC]. One of the main project is to measure the size of L1-trigger pre-firing using ECAL Trigger primitives, measure relative size of different sources and then find methods to monitor and control it in the Run 3. A number of studies have been performed in order to understand the cause and measure them. The first study includes the analysis of full readout raw data from ECAL electronics to measure the total size of pre-firing in 2017 and 2018 data, i.e. with before and after this issue was solved by resetting the online detector timing. This serves as an benchmark and all the factorized components of pre-firing should sum up to this measurement.

The initial hypothesis about the cause of pre-firing is, it is occurring due to two sources, (i) in the front end electronic chip (FENIX) and the effect is supposed to be present in the emulator and supposedly can be simulated if needed, (ii) in the trigger concentration card (TCC) where the out of phase-ness of data and TCC clock lead to shift in the time by 1 bunch crossing (BX) and lead to the pre-firing and not present in the emulator.

To measure the relative size of these two effects three additional studies are performed:

1. A study was performed using data collected in 2018 where phase between data and electronics is changed. This gives an insight on the size of pre-firing probability as a function of the time shift we might observe in future.
2. To compliment this, special data was collected to measure the effect of TCC alone, where the phase of data and TCC clock is changed manually and data is recorded and then alignment is checked. The rate of mis-alignment roughly the pre-firing probability due to the TCC alone.
3. At last the simulation setup was developed to understand the effect coming from the FENIX. As this effect is already present in the emulator and hence a shift in the timing at this stage and re-emulating the digis can provide the missing piece of information we need to understand the issue better.

Therefore, the set of four studies mentioned above gives a full picture about the ECAL Trigger pre-firing. The results are being finalised and in discussion with Level 1 trigger group to understand how study can be used in Run 3 to control the pre-firing rate. The findings and proposal for Run 3 will eventually be summarised in a detector note.

HCAL

I have actively contributed to the back-end electronics testing of Forward Hadron Calorimeter upgrade project. I set up a testing lab at SINP with help of Prof. Manoj Sharan. I was the first one to test initial prototypes of micro HCAL Trigger and Readout (μ HTR) and its various components at SINP. The power mezzanines and auxiliary power mezzanines were stress tested for long hours to ensure its quality. The task was accomplished using APIs developed in C++, with help of engineers from University of Minnesota. I have also performed the high speed links testing and characterisation of the optical splitters and Pluggable Parallel-Fiber-Optic Modules (PPODs). The backend electronics components were later installed and used for data collection during Run 2. The work is summarised in the form of an internal detector note CMS DN-2014/009.

HGCAL

I have participated in the development of the future generation calorimeter which will replace the present ECAL and HCAL in the forward direction.

- **Standalone simulation:** A standalone GEANT4 setup to simulate the proposed Particle Flow Calorimeter was developed (later named HGCAL). The initial prototype considered sampling calorimeter with gas as an active material. The measurement of resolution, linearity and Moliere radius for ECAL and HCAL were performed to optimise the geometry. The thickness of absorber layer, active material gas type and its operating pressure were main parameters used for the optimisation of the geometry. The initial studies concluded that gas based calorimeter could not meet the required for the resolution and hence Silicon was proposed and used as an active material.
- **Validation software:** Developed the software for the validation of SimHits, Digis and RecHits for HGCAL. The software packages were developed within the CMS software suite which is still being used by the collaboration to validate the new version of simulation, digitisation and reconstruction codes.
- **Test beam:** I have actively participation in data analysis of HGCAL test beam. One electronics module was tested in November 2016 test-beam with electron beam to exploit the timing capabilities of the Silicon (Si) sensors. The timing resolution of Si cells w.r.t timing reference detector was measured for each cell and then later combined with linear energy weights to achieve better performance. The study concluded that the time resolution of the Si cells can be used to separate the hard interaction and the pile up vertices in the HL-LHC phase when the average expected pileup would be 200. The results were part of the publication **JINST 13 (2018) 10, P10023**